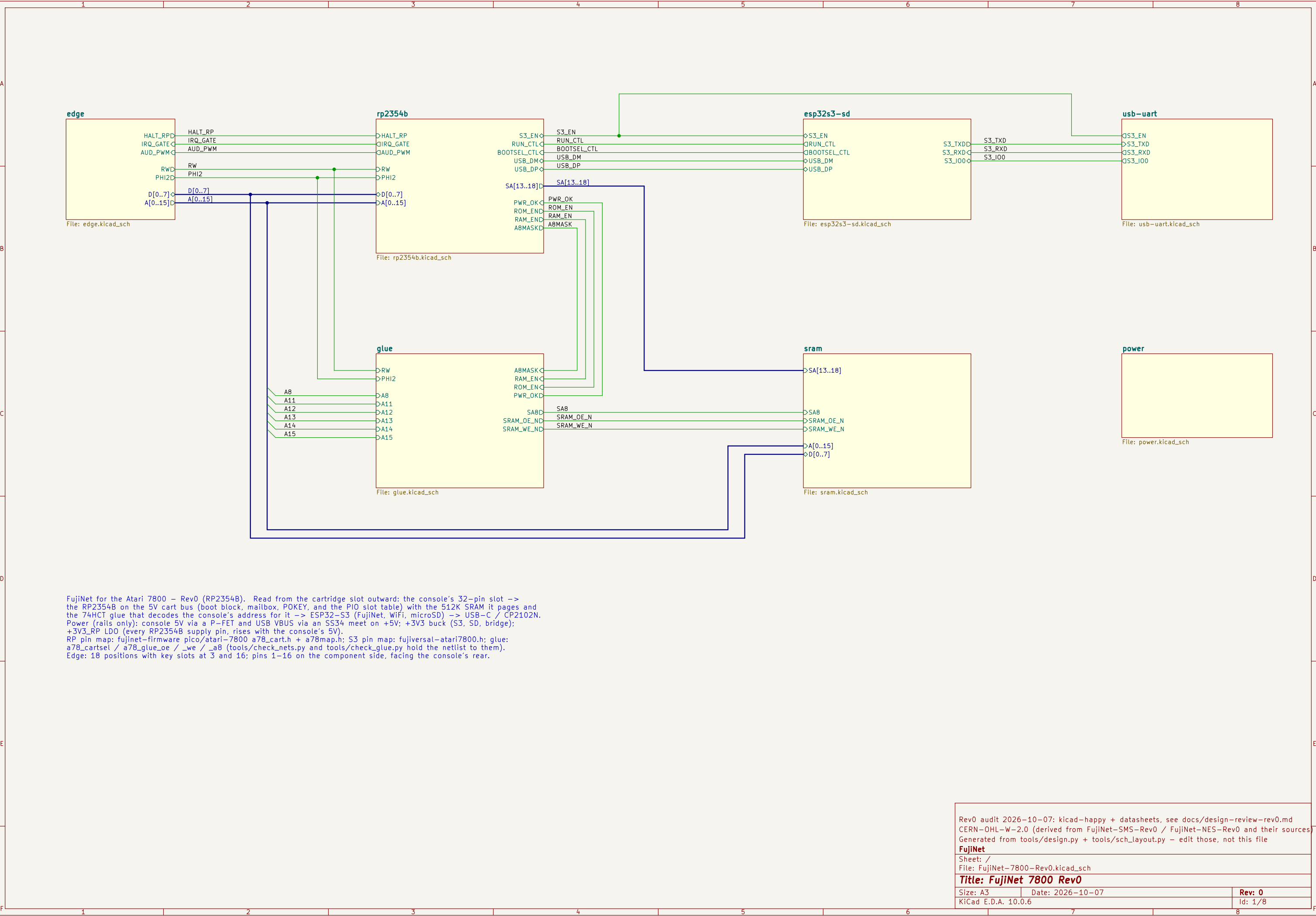
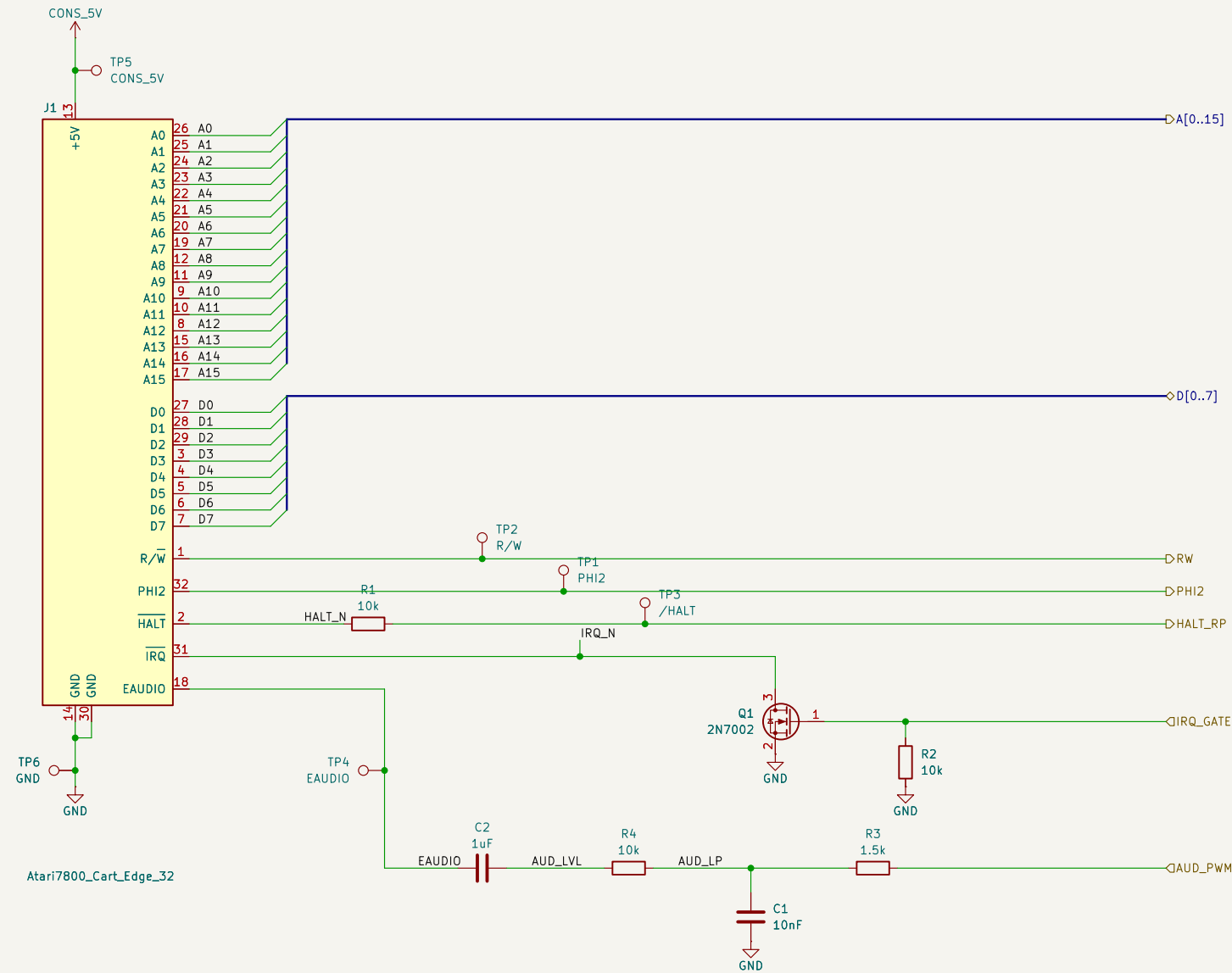
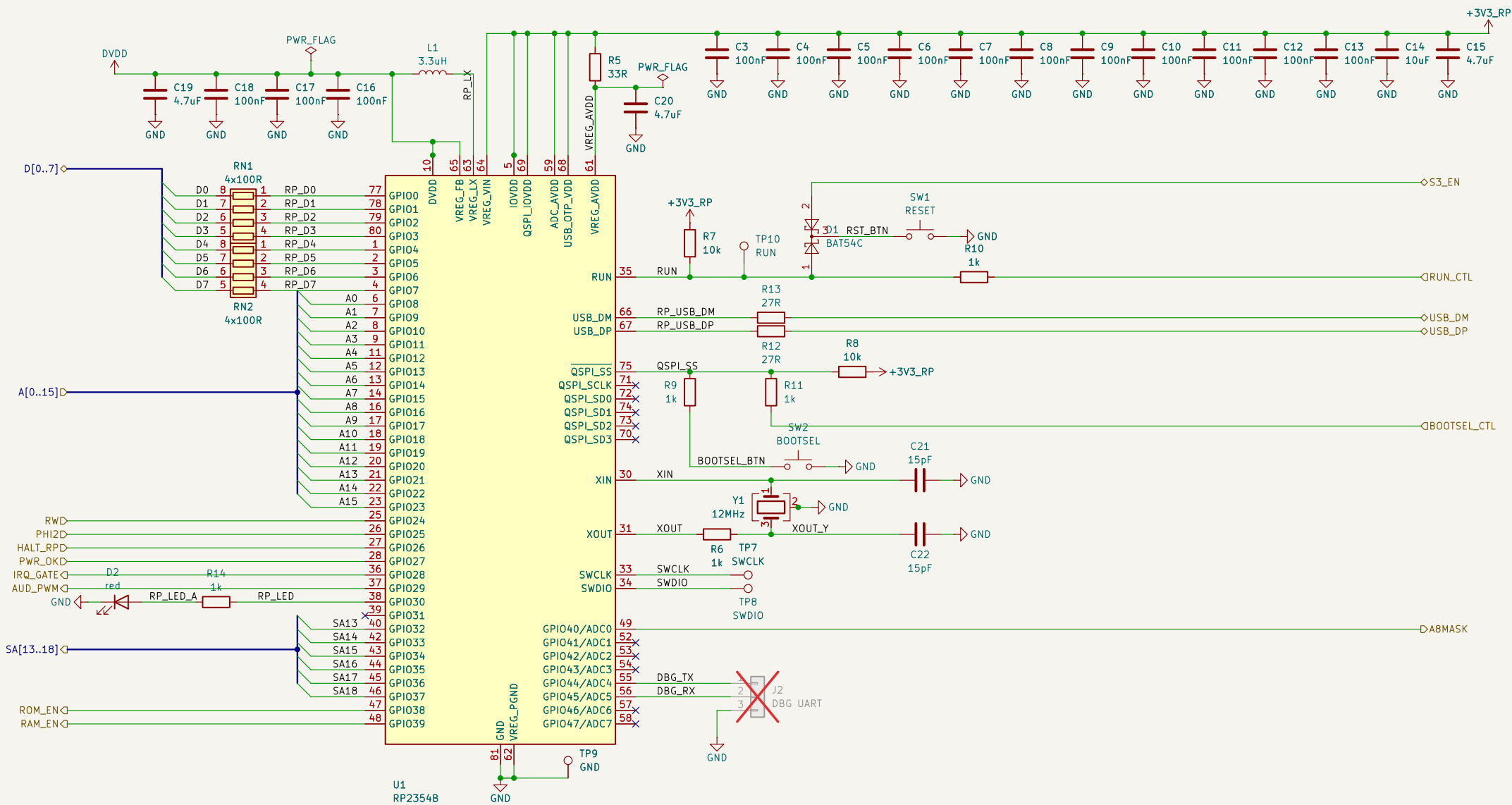




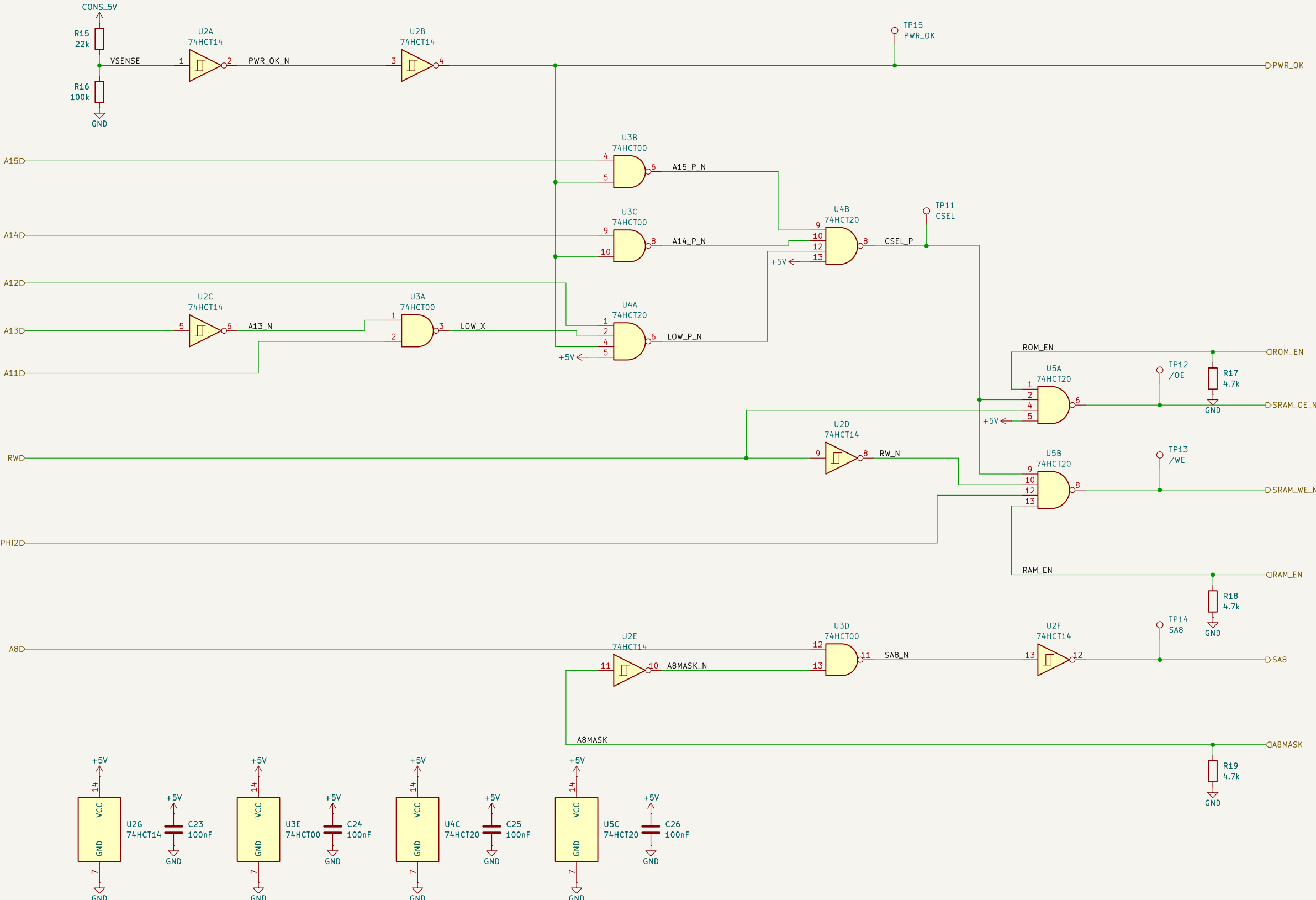
FujiNet for the Atari 7800 – Rev0 (RP2354B). Read from the cartridge slot outward: the console’s 32-pin slot –> the RP2354B on the 5V cart bus (boot block, mailbox, POKEY, and the PIO slot table) with the 512K SRAM it pages and the 74HCT glue that decodes the console’s address for it –> ESP32–S3 (FujiNet, WiFi, microSD) –> USB–C / CP2102N. Power (rails only): console 5V via a P–FET and USB VBUS via an S534 meet on +5V; +3V3 buck (S3, SD, bridge); +3V3_RP LDO (every RP2354B supply pin, rises with the console’s 5V). RP pin map: fujinet–firmware pico/atari–7800 a78_cart.h + a78map.h; S3 pin map: fujiversal–atari7800.h; glue: a78_cartsel / a78_glue_oe / _we / _a8 (tools/check_nets.py and tools/check_glue.py hold the netlist to them). Edge: 18 positions with key slots at 3 and 16; pins 1–16 on the component side, facing the console’s rear.

Atari 7800 32-pin cartridge slot, drawn by function (finger numbers on the pins). 18 positions at 2.54 mm per face; positions 3 and 16 are key slots in the board for the console connector's plastic keys. Pins 1–16 on F.Cu, the component side, which faces the console's REAR; pin 1 at the left seen from F.Cu, fingers down; pin k over pin 33–k on B.Cu (tools/edge_geom.py, tools/audit/edge_orientation.py). No chip select and no reset on this edge; the glue decodes A15–A11. Cart audio: GPIO29 PWM (833 kHz carrier) → 1.5k / 10n (fc 10.6 kHz) → 10k level → 1u → EAUDIO; the console adds 0.1u + 6.8k into its audio sum (C10, R5), as a POKEY cart's 1k pull-up + 12k does. /IRQ: the 2N7002 pulls it low only while GPIO28 is high (the firmware never does); console pull-up R32 2.2k. /HALT (MARIA's weak MOS output) through 10k to the RP: observed only. While the console's BIOS is mapped its U4 (74LS08) holds A15, A14, A12 low at this edge.

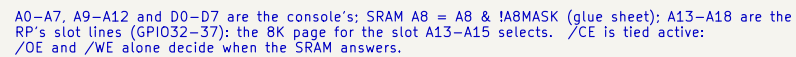




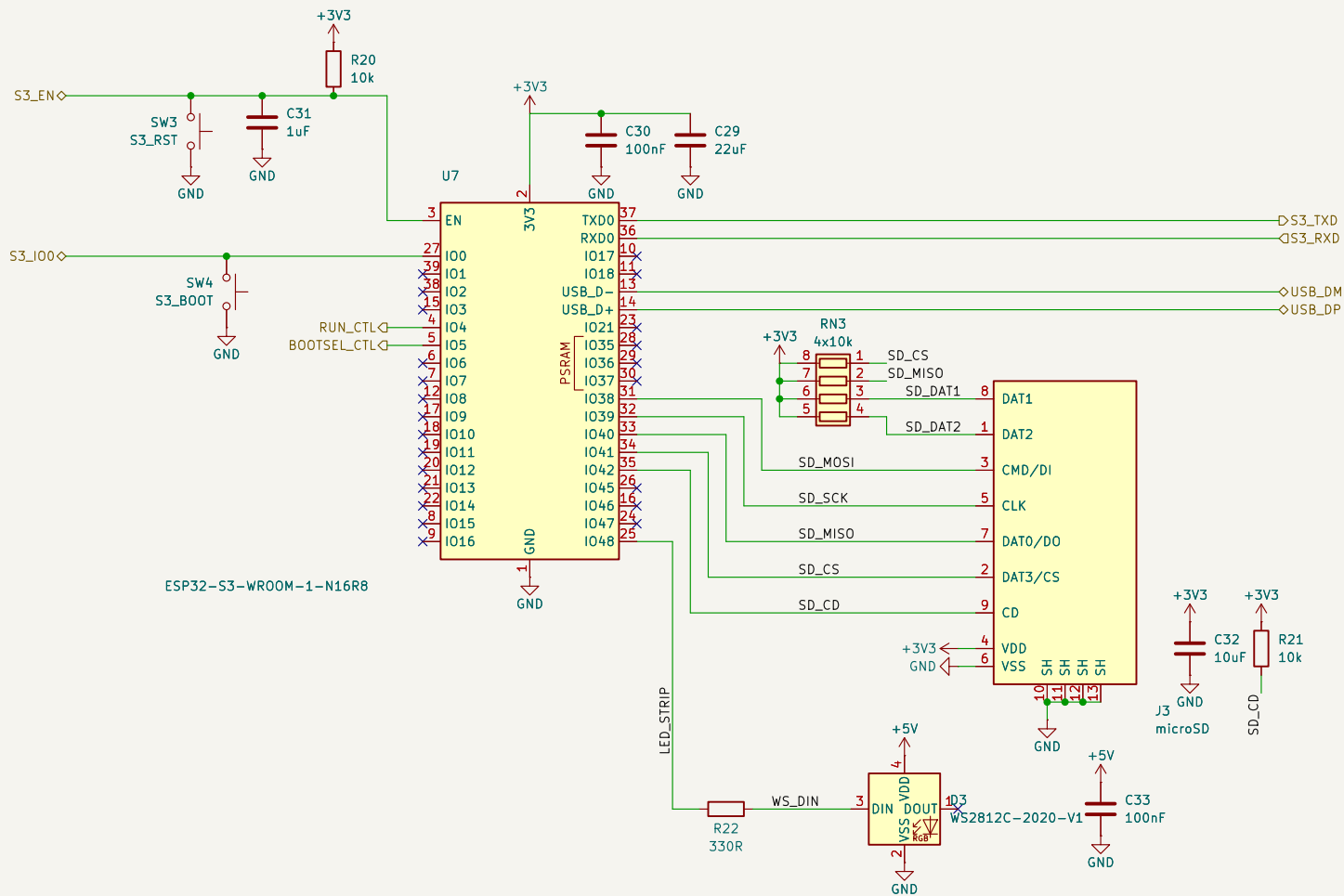
fujinet-firmware pico/atari-7800 a78_cart.h: D0-D7 GPIO0-7 (through 100R), A0-A15 GPIO8-23, R/W 24, PHI2 25, /HALT 26 (observed only), PWR_OK 27, /IRQ gate 28, audio PWM 29, LED 30: a78map.h slot word on GPIO32-40 (PIO, indexed by A13-A15): SRAM A13-A18, ROM_EN, RAM_EN, A8MASK. GPIO40-47 are not 5 V tolerant: A8MASK drives a 74HCT14 input only, GPIO44/45 the 3.3 V debug header (DNP). One 100 nF per IOVDD pin (key C_IOV<pin>: placed at that pin). RP2350 datasheet 6.3.7: VREG_VIN and IOVDD on one 3.3 V supply (+3V3_RP, the LDO that rises with the console 5V). VREG_AVDD through 33R / 4.7 uF, 3.3 uH from VREG_LX to DVDD (= VREG_FB).



fujinet-firmware pico/atari-7800 a78_cart.h: a78_cartsel(), a78_glue_oe(), a78_glue_we(), a78_glue_a8().
RW = 1 for a read; ROM_EN, RAM_EN, A8MASK = RP GPIO38, 39, 40 (the PIO slot table; 4.7k pull-downs hold them low until it runs, RP2350-E9); PWR_OK = console +5V present (0.82 x CONS_5V into the 74HCT14).
CARTSEL = A15 | A14 | A12 & (A13 | !A11) (\$4000-\$FFFF, \$1000-\$17FF, \$3000-\$3FFF)
CSEL_P = PWR_OK & CARTSEL = NAND(!A15 & PWR_OK), !A14 & PWR_OK, !A12 & LOW_X & PWR_OK))
SRAM /OE = !(CSEL_P & RW & ROM_EN) (no PHI2; MARIA DMA is not PHI2-aligned)
SRAM /WE = !(CSEL_P & !RW & PHI2 & RAM_EN) (PHI2 one gate from /WE); SRAM A8 = A8 & !A8MASK
tools/check_glue.py evaluates this netlist against the firmware's C for every input combination.



Id: 5/8



Rev0 audit 2026-10-07: kicad-happy + datasheets, see docs/design-review-rev0.md
 CERN-OHL-W-2.0 (derived from FujiNet-SMS-Rev0 / FujiNet-NES-Rev0 and their sources)
 Generated from tools/design.py + tools/sch_layout.py - edit those, not this file

FujiNet

Sheet: /esp32s3-sd/
 File: esp32s3-sd.kicad_sch

Title: FujiNet Rev0 - ESP32-S3 FujiNet, microSD, LED

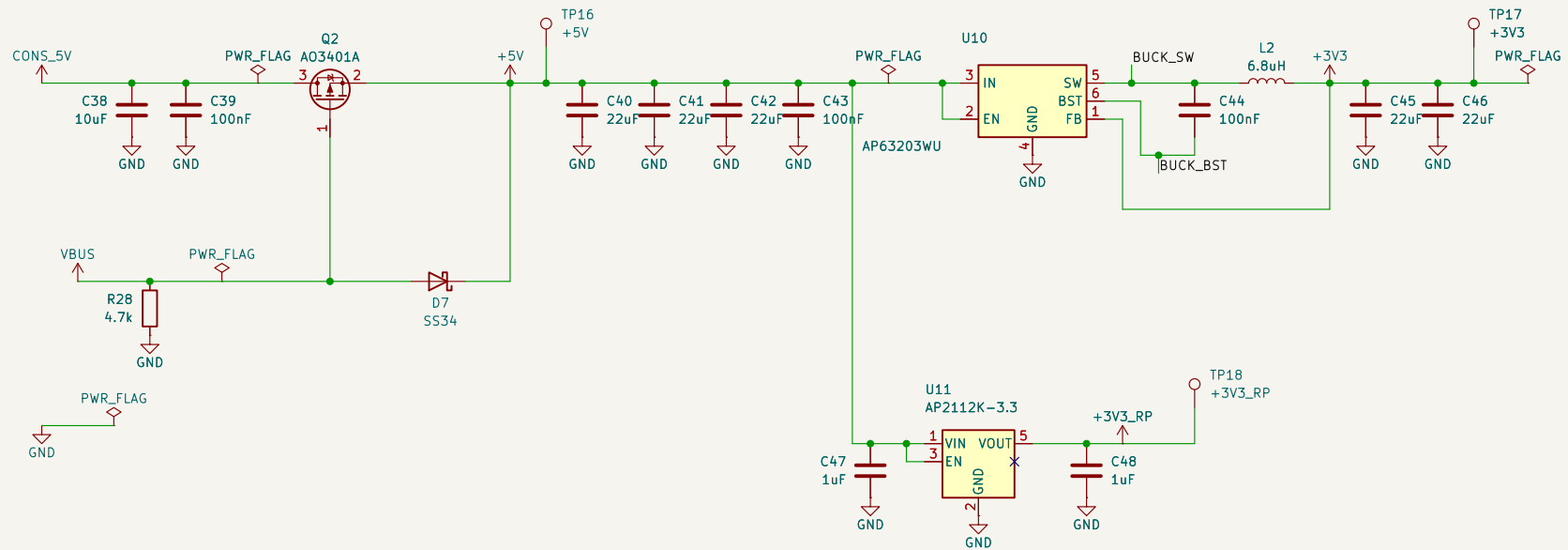
Size: A4 Date: 2026-10-07

Rev: 0

KiCad E.D.A. 10.0.6

Id: 6/8

Id: 7/8



Console 5V (slot pin 13, CONS_5V) through the A03401A (gate = VBUS: fully on without USB, body diode only with USB, so nothing back-feeds the console); USB VBUS through the SS34; R_VBPD 4.7k holds the gate low against the SS34's reverse leakage. +5V: SRAM, glue, WS2812, buck, RP LDO. +3V3 (buck): S3, microSD, CP2102N. +3V3_RP (LDO, rises with the 5V rail): every RP2354B supply pin, so its 5V-tolerant pads are powered before the bus is.

Rev0 audit 2026-10-07: kicad-happy + datasheets, see docs/design-review-rev0.md
CERN-OHL-W-2.0 (derived from FujiNet-SMS-Rev0 / FujiNet-NES-Rev0 and their sources)
Generated from tools/design.py + tools/sch_layout.py - edit those, not this file

FujiNet

Sheet: /power/
File: power.kicad_sch

Title: FujiNet 7800 Rev0 - Power: 5V OR, 3.3V buck, RP LDO

Size: A4 Date: 2026-10-07

KiCad E.D.A. 10.0.6

Rev: 0

Id: 8/8